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EMPATHY, KEY TO QUICK HEALING, SOMETHING AI CANNOT DELIVER

No matter the industry, artificial intelligence (AI) has become commonplace. When it comes to medicine, AI helps surgeons to streamline tasks, improve operational efficiencies and simplify complex procedures.

Despite its potential to unlock new insights and streamline the way providers and patients interact with healthcare data, AI may bring considerable threats of privacy, ethical concerns, and medical errors. In today's era, AI is about the new doctor and how they find their place in the artificial intelligence-enabled health system. The need is of a human–artificial intelligence interface, the knowledge-handler, the empathic communicator.

Informs Dr. Ramneek Mahajan, Senior Director and Head Joint Reconstruction Unit (Knee & Hip), Max Smart Super Specialty Hospital, Saket, New Delhi "In orthopedics, AI holds promise for several cutting-edge applications that can transform the quality of care rendered, accelerate the delivery of services, and improve the value rendered." Although AI has come a long way in the medical world, human surveillance is still essential. For example, surgery robots operate logically, as opposed to empathetically.

Surgeons may notice vital behavioral observations that can help diagnose or prevent medical complications. AI can identify potential ailments faster than human doctors, but when it comes to decision-making, AI cannot yet fully and safely take over for human physicians. AI cannot adapt to the variety of situations arising during a difficult surgery as it has a pre-fixed analytic but a surgeon can not only adapt to the situation, but find new ways on the operating table to save a person's life.

A good surgeon's job requires creativity and problem-solving abilities, something that robots and algorithms will never have. AI uses a neural network and utilizes the same algorithm for all patient data but medicine as a branch has always focused and prided itself on individualized patient treatment. There are many kinds of patients, each with different lifestyles. Therefore, no two cases are the same, each one is entirely different and requires human physicians.

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Informs Dr. Ramneek Mahajan, "Robots can't show empathy. It is one of the key elements of quality healthcare. It improves patient satisfaction and promotes healing. Unfortunately, empathy is unachievable for an automated machine." We might teach robots to mimic these things, but sincerity can't be taught.

Despite the fact that robots are in many ways seen as more accurate and reliable than human surgeons, people are inevitably worried about putting their lives in the hands of a machine. Fears surrounding the safety and accuracy of procedures are prevalent among members of the public, however, it is highly likely that the widespread use of medical robotics is set to continue.

While there are innumerable benefits of employing robots, there are chances of errors and failures. As in all newer endeavors, AI is the answer only when the person behind the machine asks the right question. AI is a new tool and can perform more complicated tasks and detect minor errors which can be missed by humans, but there are costs and risks if it is utilized incorrectly. There is always some scope for human error or mechanical failure with these advanced robots. A single mechanical malfunction can cost human lives. Informs Dr. Ramneek Mahajan, "In robotic joint replacement, the surgeon is forced to use the prosthesis of the same company of the robot as it is locked. Knowing that the prosthesis lacks sufficient clinical data and hence not be suitable for the patient in the long run, the surgeon compromises on it and uses it." Medical AI depends heavily on diagnosis data available from millions of catalogued cases. In cases where little data exists on particular illnesses, demographics, or environmental factors, a misdiagnosis is entirely possible.

As technology advances further in the world around us, the role that AI applications play in orthopedics is already seen by many as an inevitable future step. Much investment and early work have already explored the optimization of early utilization of such approaches, across a range of clinical and para-clinical domains. Like any new technology, effort must be expended to ensure AI applications touted for clinical use have shown evidence-based rationales for adoption with non-inferiority (but ideally improved) outcomes against the conventional standards.

It is important that the science underpinning such advancements is not outpaced by the hype. While it is likely that computer- and AI-based programs will add value to areas where human cognition and capacity stand as rate-limiting factors, the expense and effort required to establish such systems must be positively weighed against the perceivable benefit. Being such a novel and unprecedented frontier, AI in medicine presently lacks widely-accepted governance and regulatory provisions which will need to evolve at a similar rate to ensure safe and optimal utilization with respect for individual patient data and circumstances